OFFICIAL

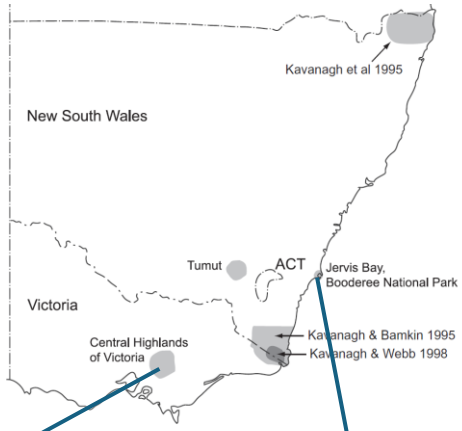
NRC Project B9:
Historical trends in greater glider populations on state forests

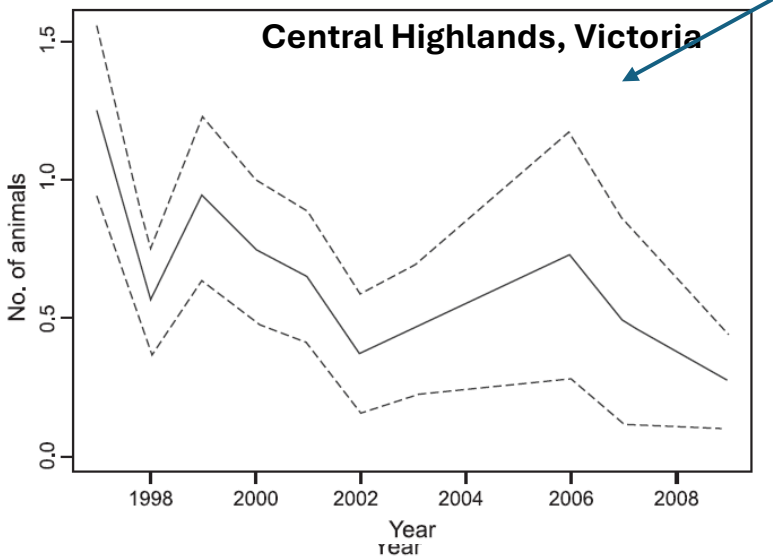
Modelling the effects of climate, logging, fire, and other environmental covariates on the occupancy and abundance of the southern greater glider (*Petauroides volans*) in the Coastal IFOA landscape

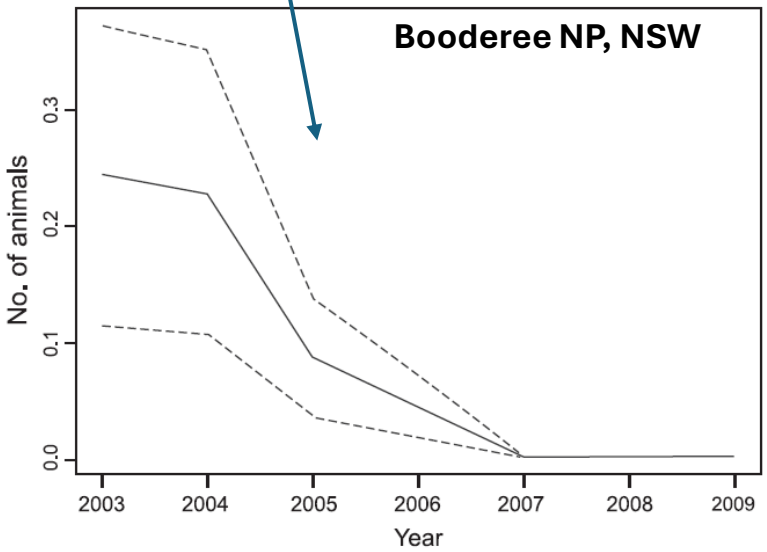
Prof. Craig Nitschke
Dr. Benjamin Wagner
Dr. Raphael Trouvé
Dr. Rod Kavanagh



Population Trends



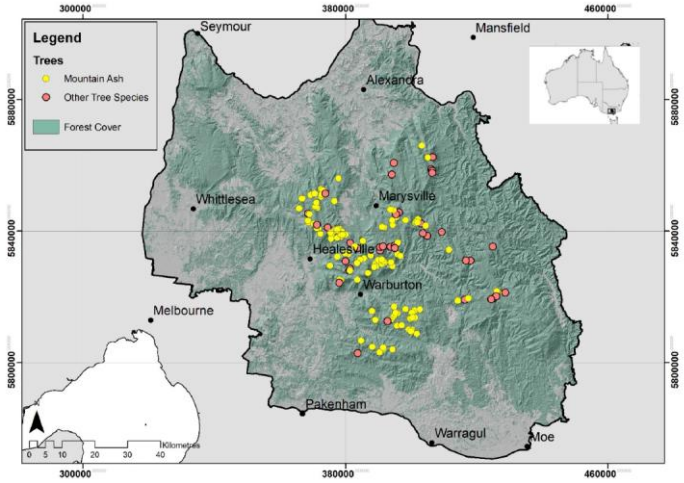


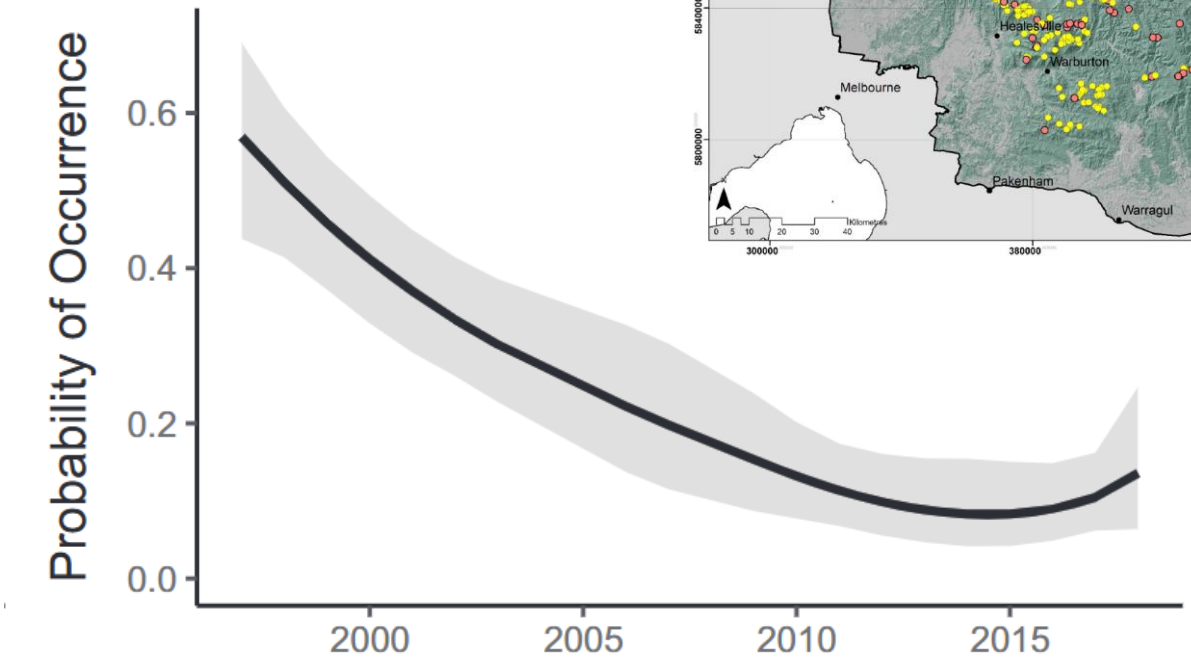


Lindenmayer et al. 2011



Population Trends





Lindenmayer et al. 2021



OFFICIAL

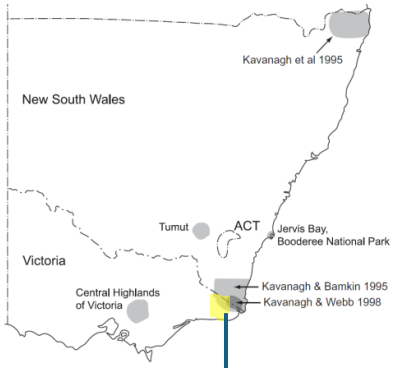
esa

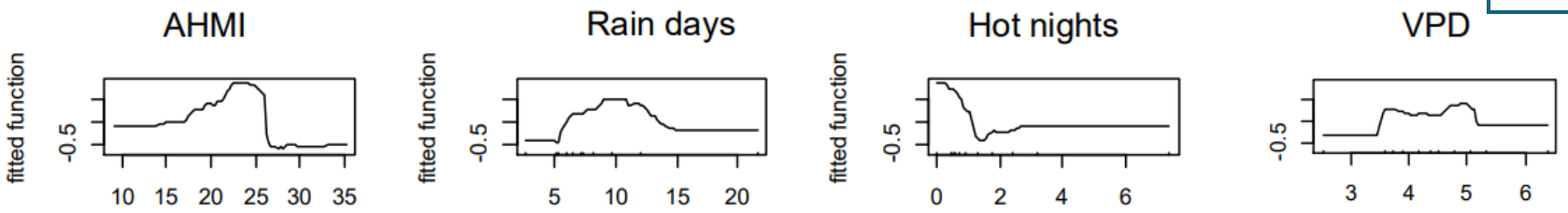
ECOSPHERE

Climate change drives habitat contraction of a nocturnal arboreal marsupial at its physiological limits

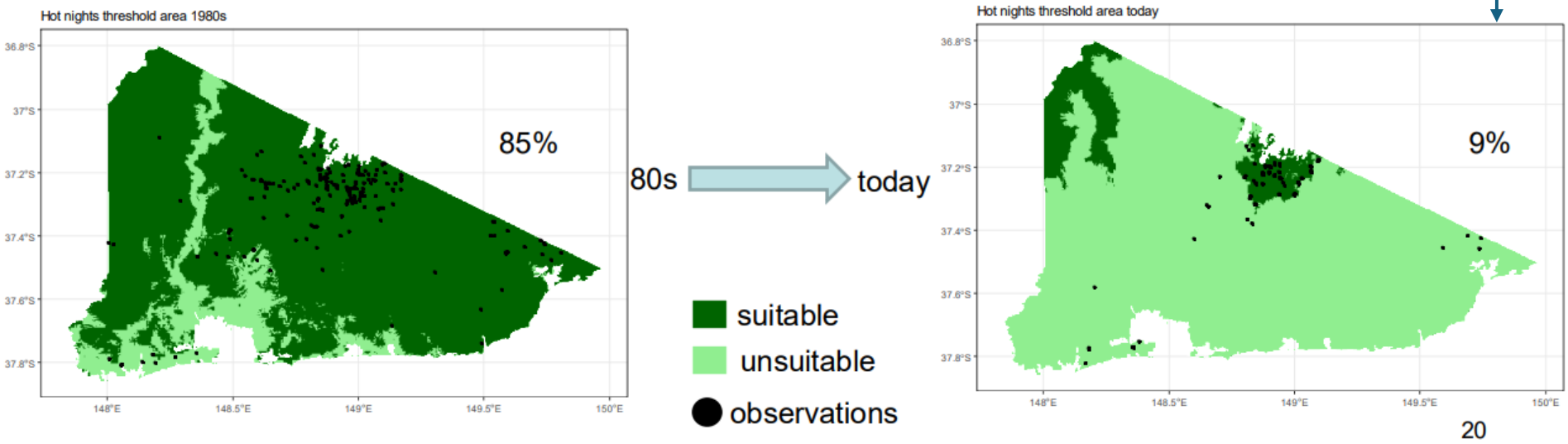
BENJAMIN WAGNER,[1,†] PATRICK J. BAKER,[1] STEPHEN B. STEWART,[1,2] LINDA F. LUMSDEN,[3] JENNY L. NELSON,[3] JEMMA K. CRIPPS,[3] LOUISE K. DURKIN,[3] MICHAEL P. SCROGGIE,[3,4] AND CRAIG R. NITSCHKE[1]







Thresholds: 14 - 25 >100 per decade <50 per decade 3.5 - 5



OFFICIAL

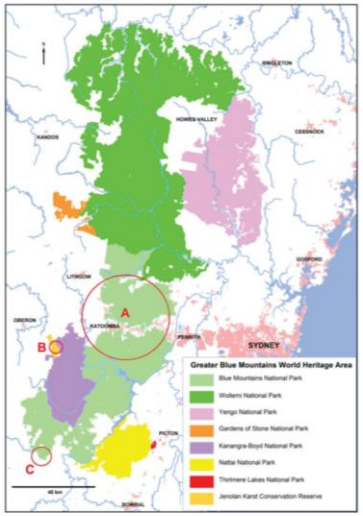
Impact of the 2019-20 drought, heatwaves and mega-fires on Greater Gliders (*Petauroides volans*) in the Greater Blue Mountains World Heritage Area, New South Wales

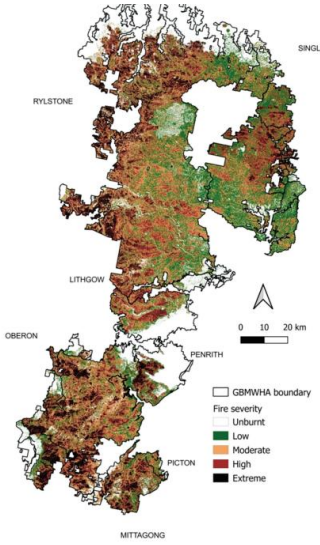
Peter Smith and Judy Smith 2022



Figure 1. The Greater Glider is highly variable in colour, with a dark morph (A) and a less common pale morph (B). The two morphs often occur together and intermediate forms also occur (the dark animal shown has some intermediate features).

2022 Australian Zoologist volume 42 (2) 165





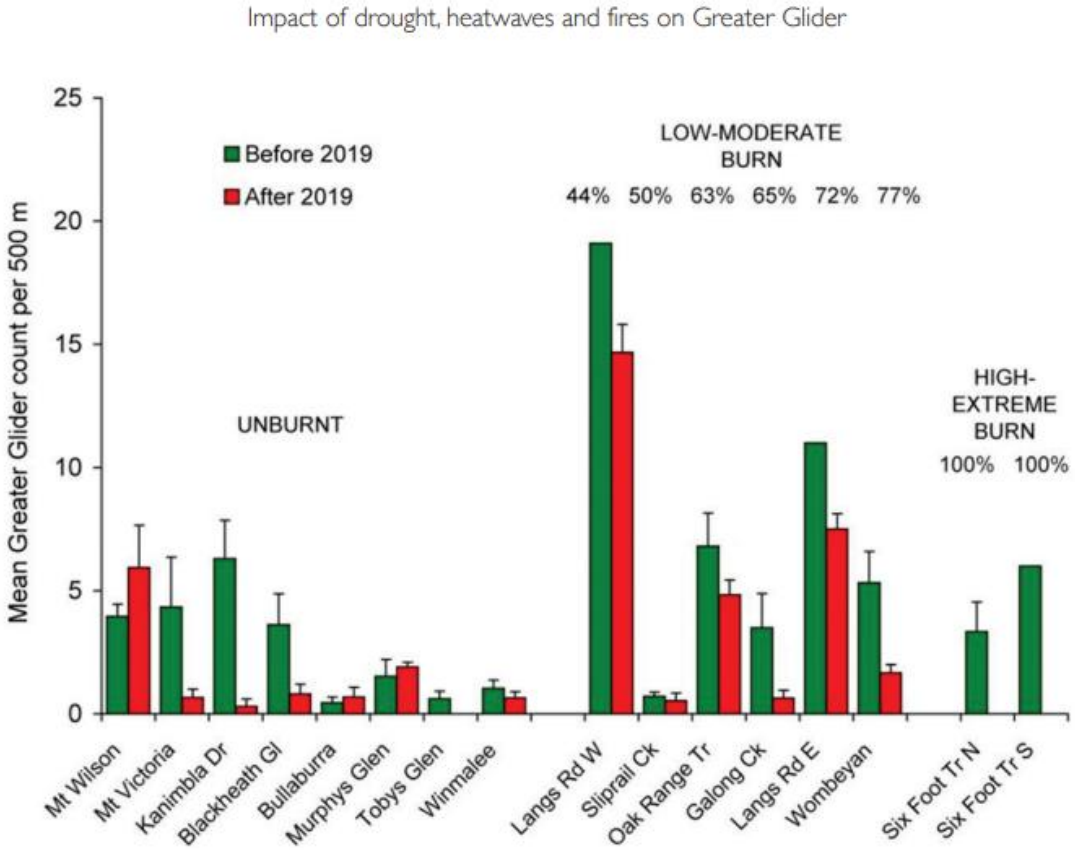


Figure 5. Comparison of mean Greater Glider counts before and after the drought, heatwaves and fires of 2 Figures above the columns are the % of eucalypt foliage killed by the fire. Error bars are standard errors. C without error bars are single counts.

OFFICIAL

OFFICIAL

CSIRO PUBLISHING

RESEARCH PAPER
https://doi.org/10.1071/PC24061

PACIFIC CONSERVATION BIOLOGY

Recovery of the Southern Greater Glider (*Petauroides volans*) following the extreme drought, heatwaves, and mega-fires of 2019–2020 in the southern Greater Blue Mountains World Heritage Area, Australia

Peter Smith[A,*] and Judy Smith[A]

2025



Fig. 1. Images of the Southern Greater Glider (*Petauroides volans*) from this study. (*a*) Dark phase, Transect 5, May 2024 and (*b*) pale phase, Transect 12, May 2023.

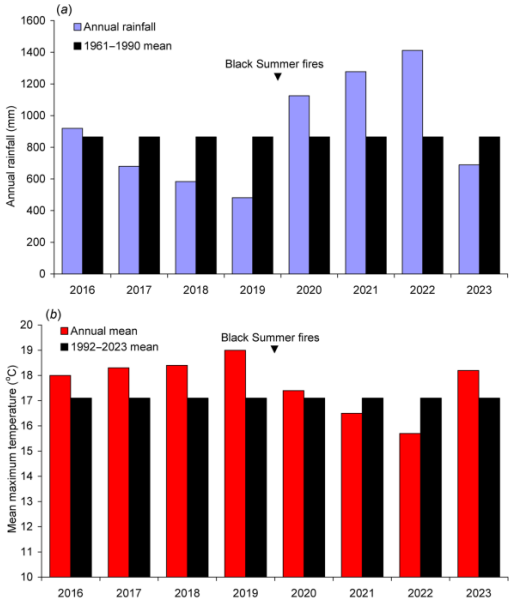


Fig. 3. (*a*) Annual rainfall and (*b*) annual mean maximum temperature in the years before and after the 2019–2020 fires. Australian Bureau of Meteorology records from Wombeyan Caves (rainfall) and Mount Boyce (temperature).

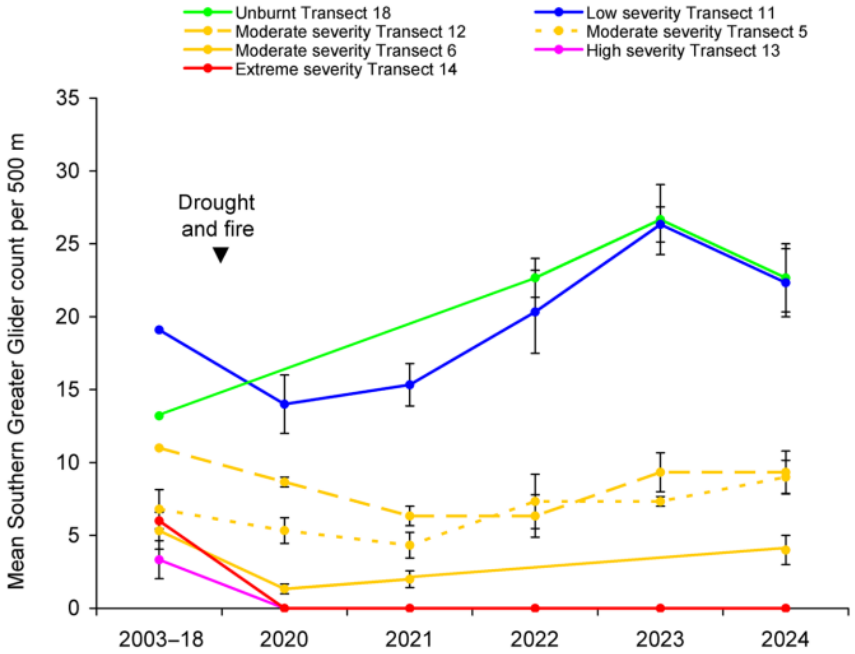
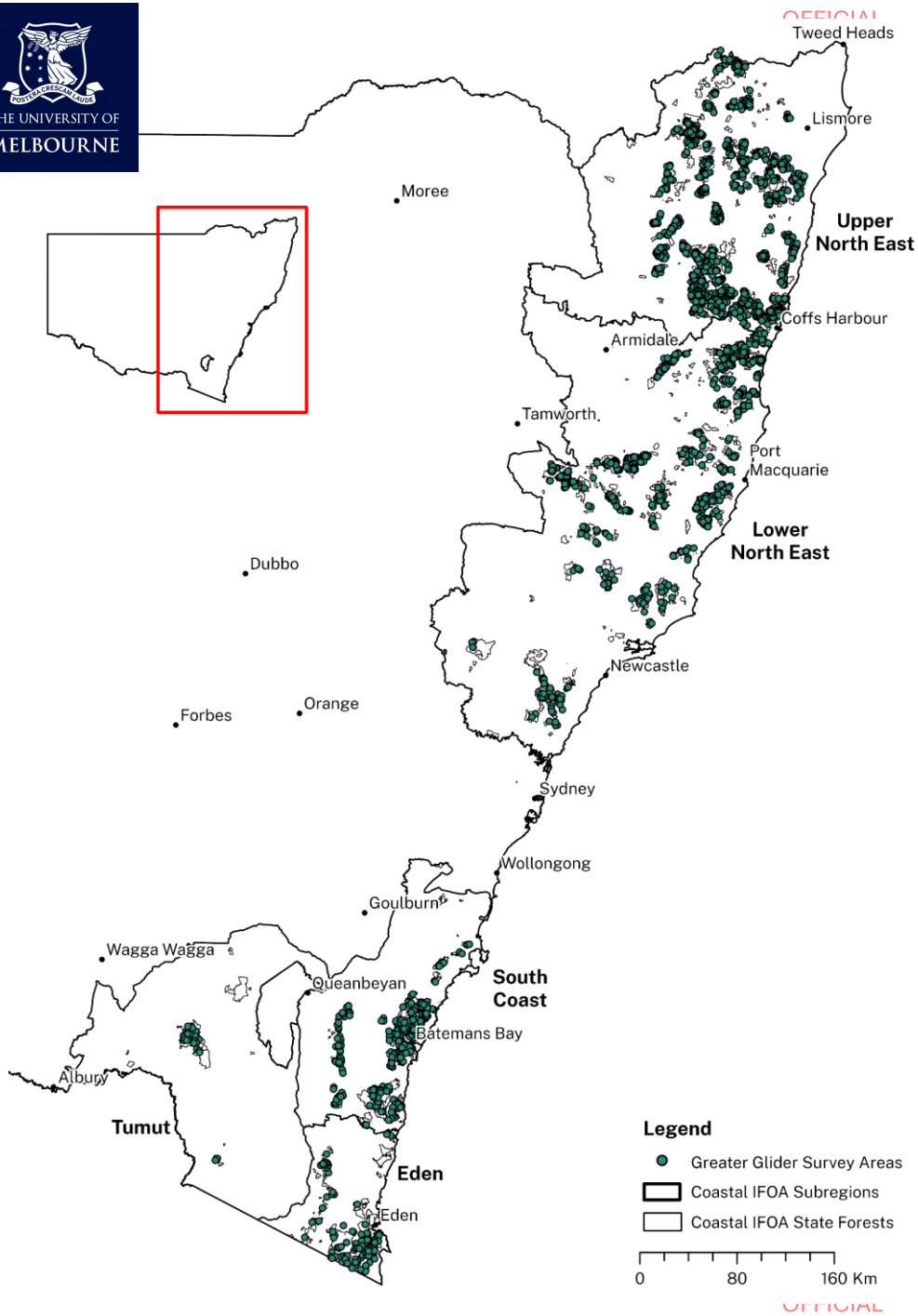
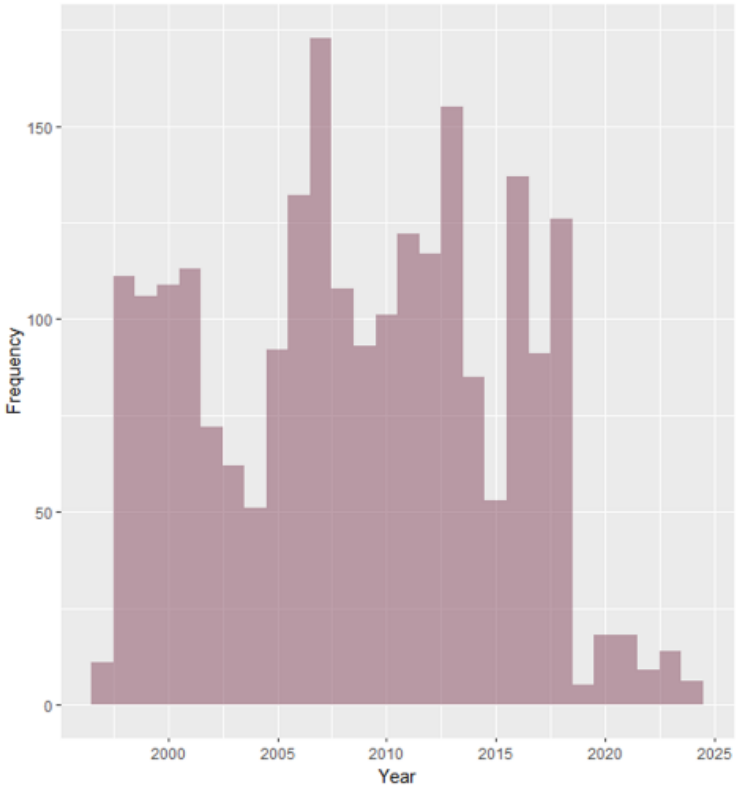


Fig. 4. Southern Greater Glider survey results for the seven 500-m transects at Wombeyan and Jenolan. Error bars, s.e. Three counts per point except for single counts in Transects 11, 12, 14, and 18 pre-2019, and four counts in Transects 5 and 6 pre-2019.

OFFICIAL





n = 2290

Years:
1997-2024

Journal of Statistical Software

Information on Mission | Information for Authors | Style Guide | Volumes | About

Home / Archives / Vol. 43 (2011) / Issue 10

unmarked: An R Package for Fitting Hierarchical Models of Wildlife Occurrence and Abundance

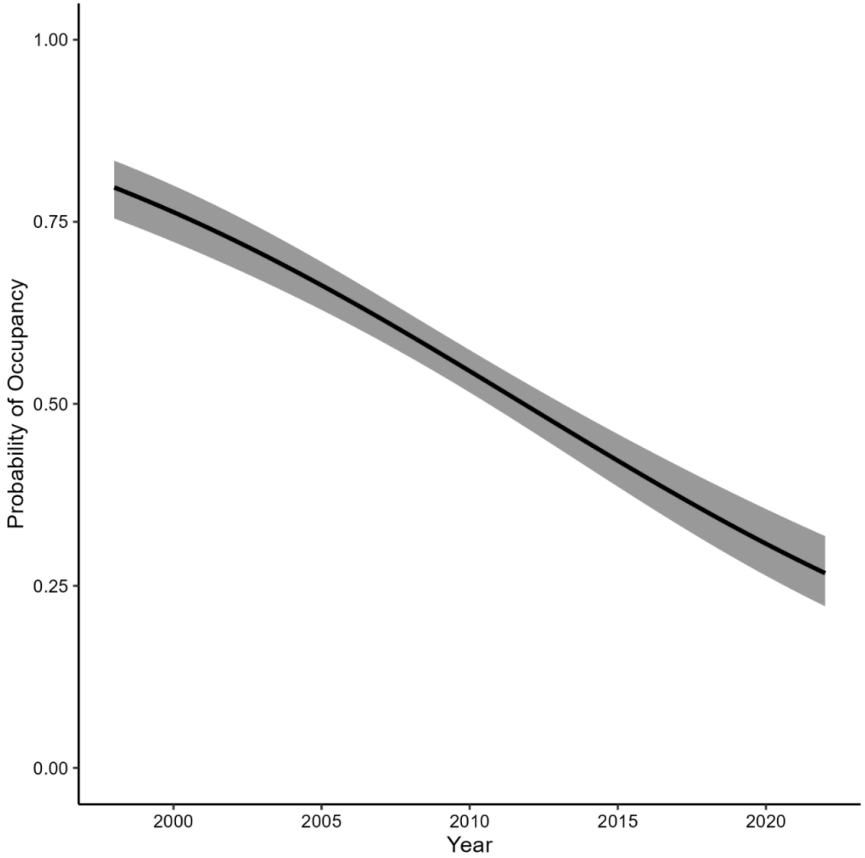
Ian Fiske, Richard Chandler

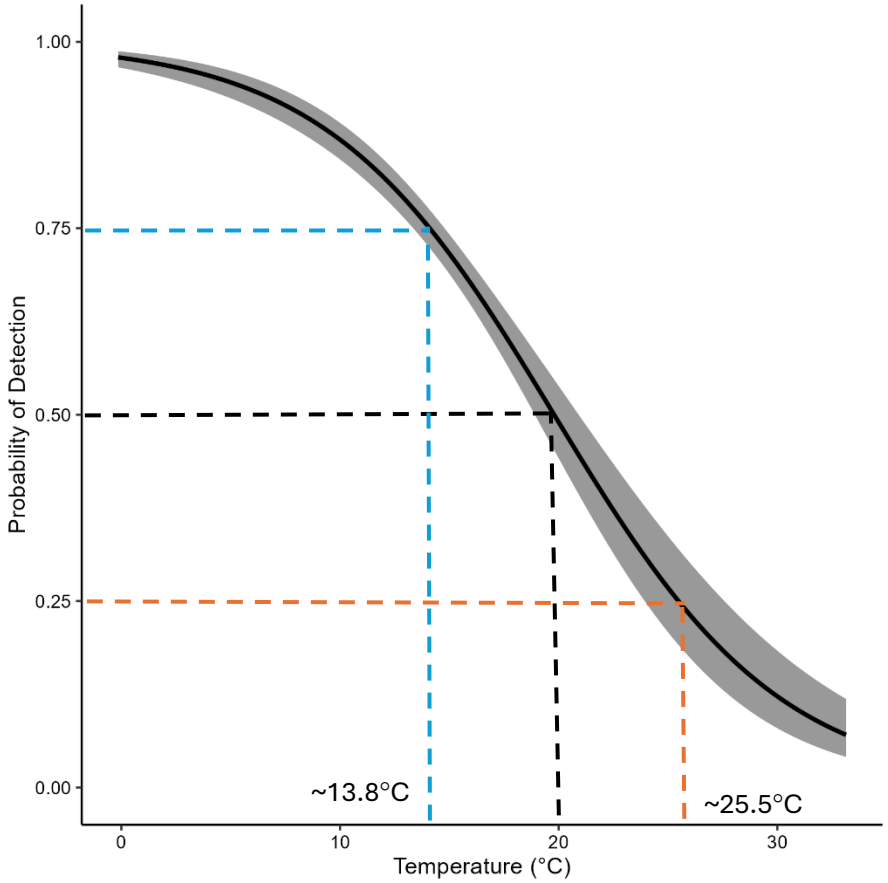
N-mixture models were used to estimate the occupancy of the species at each site using a Bernoulli distribution while considering detectability.

N-mixture models were used to estimate abundance of the species at each site using a Bernoulli distribution to account for detectability and a Poisson distribution to model observed counts of SGG.

Part I: Occupancy and Detection Modelling within dataset

Detection



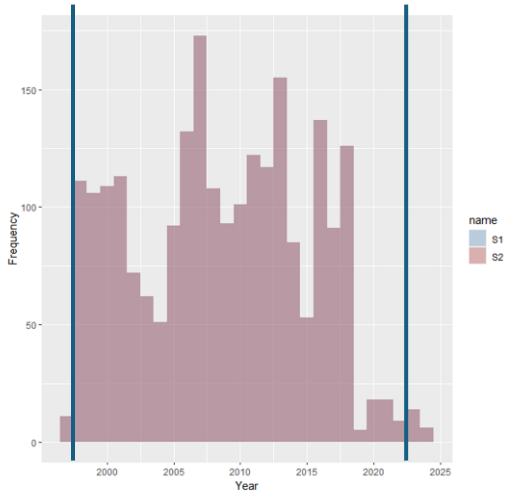


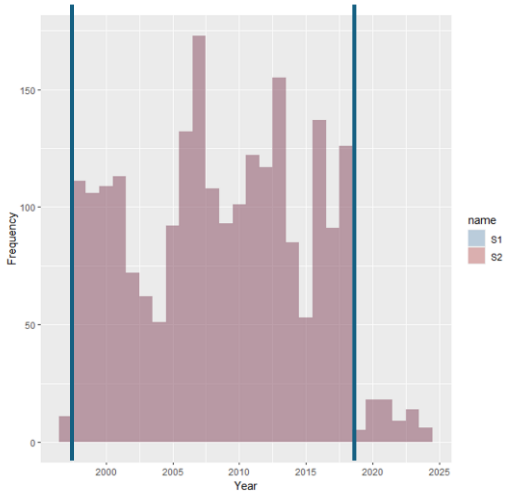
The occupancy rate was 52.5% and the detection rate was 75.8%

Part II:

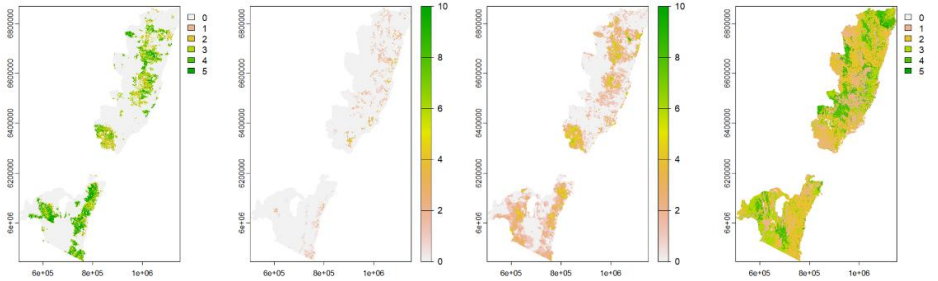
The role of climate, fire, historical harvesting and other environmental factors on SGG Occupancy and Abundance

1998-2022

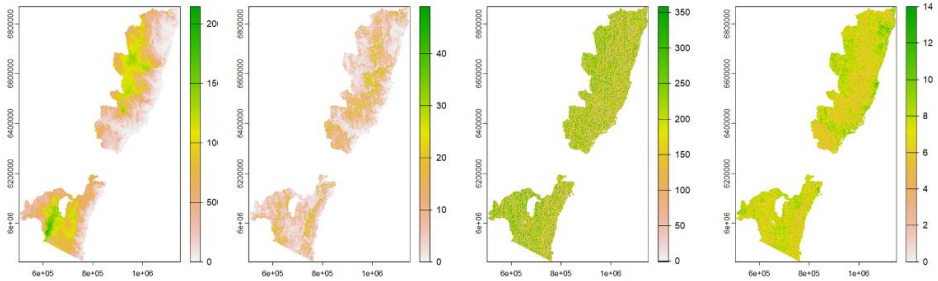




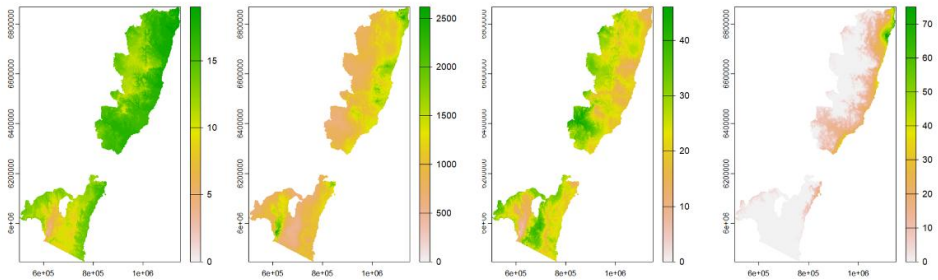
1998-2018



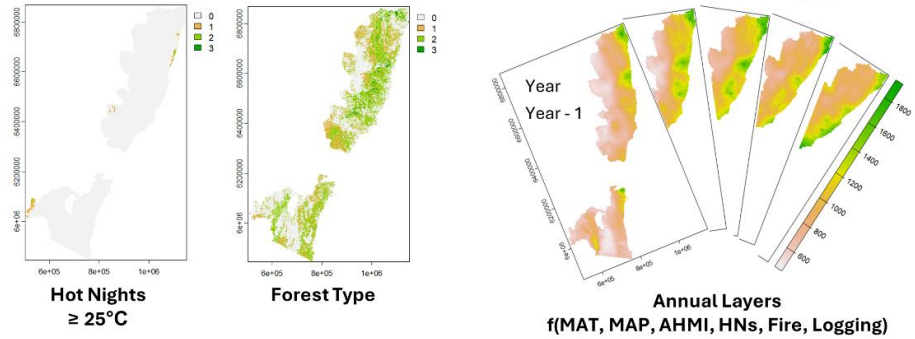
2019-2020 Fire Severity | Logging Frequency | Fire Frequency | Soil Fertility



Elevation (m) | Slope (°) | Aspect (°) | TWI

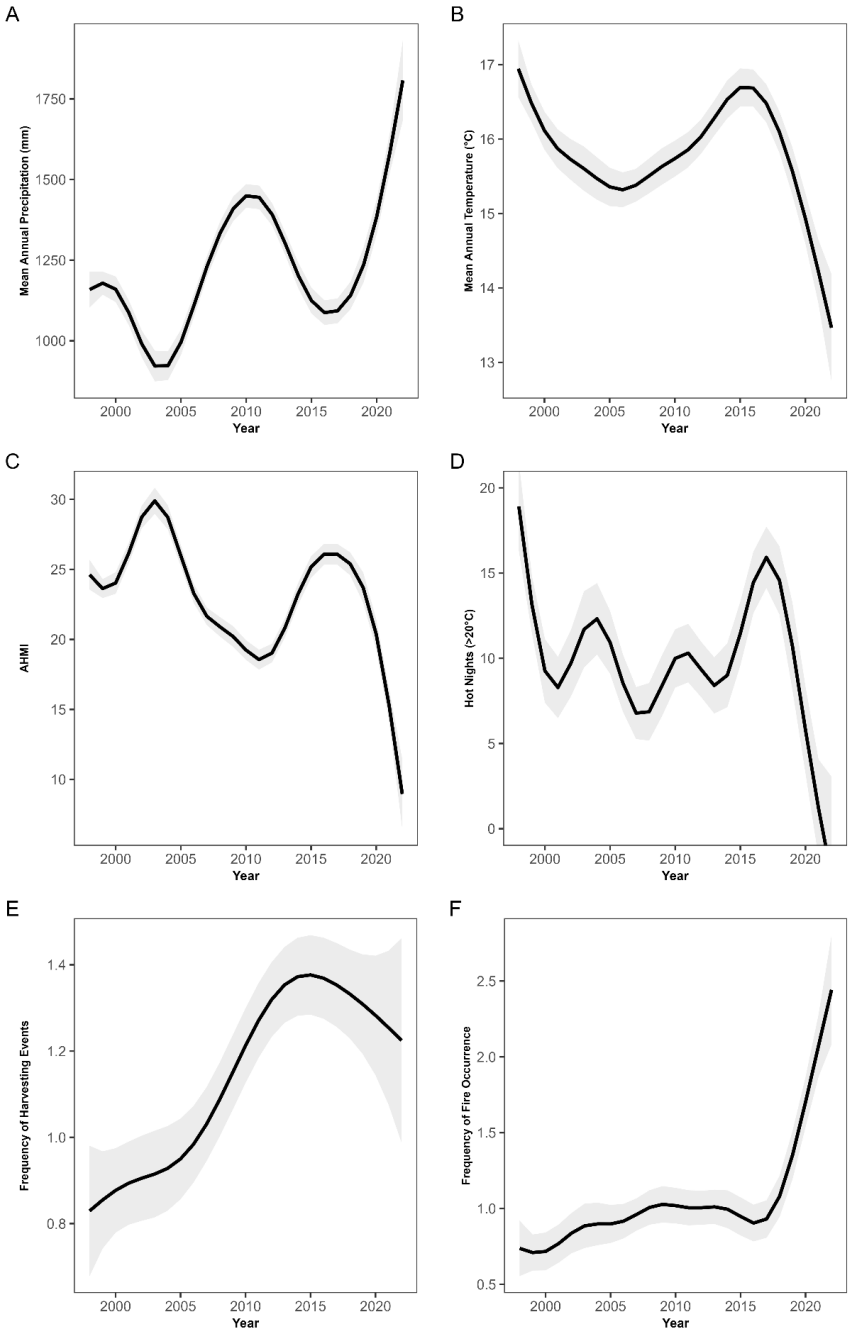


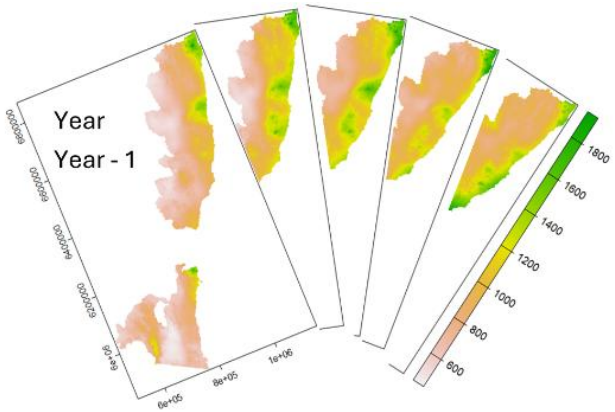
MAT (°C) | MAP (mm) | AHMI | Hot Nights (HN) ≥ 20°C



Hot Nights ≥ 25°C | Forest Type | Annual Layers f(MAT, MAP, AHMI, HNs, Fire, Logging)







Annual Layers
f(MAT, MAP, AHMI, HNs, Fire, Logging)

Trends in key covariates over time across CIFOA study area:

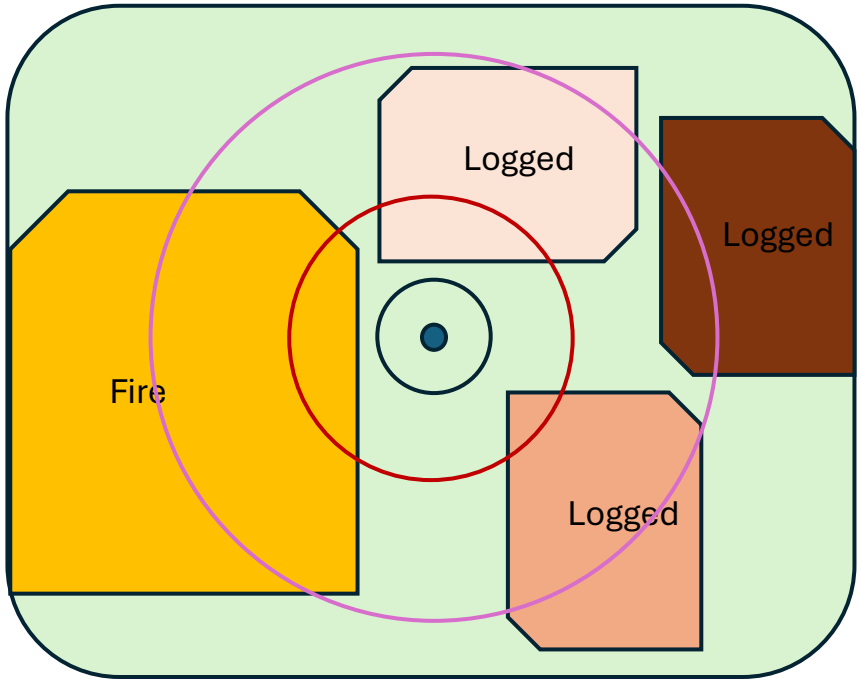
(a) Mean Annual Precipitation (MAP),
(b) Mean Annual Temperature (MAT),
(c) AHMI (aridity),
(d) Number of hot nights (HN) > 20°C,
(e) Mean logging frequency; and
(f) Mean fire frequency.

Spatial Buffering

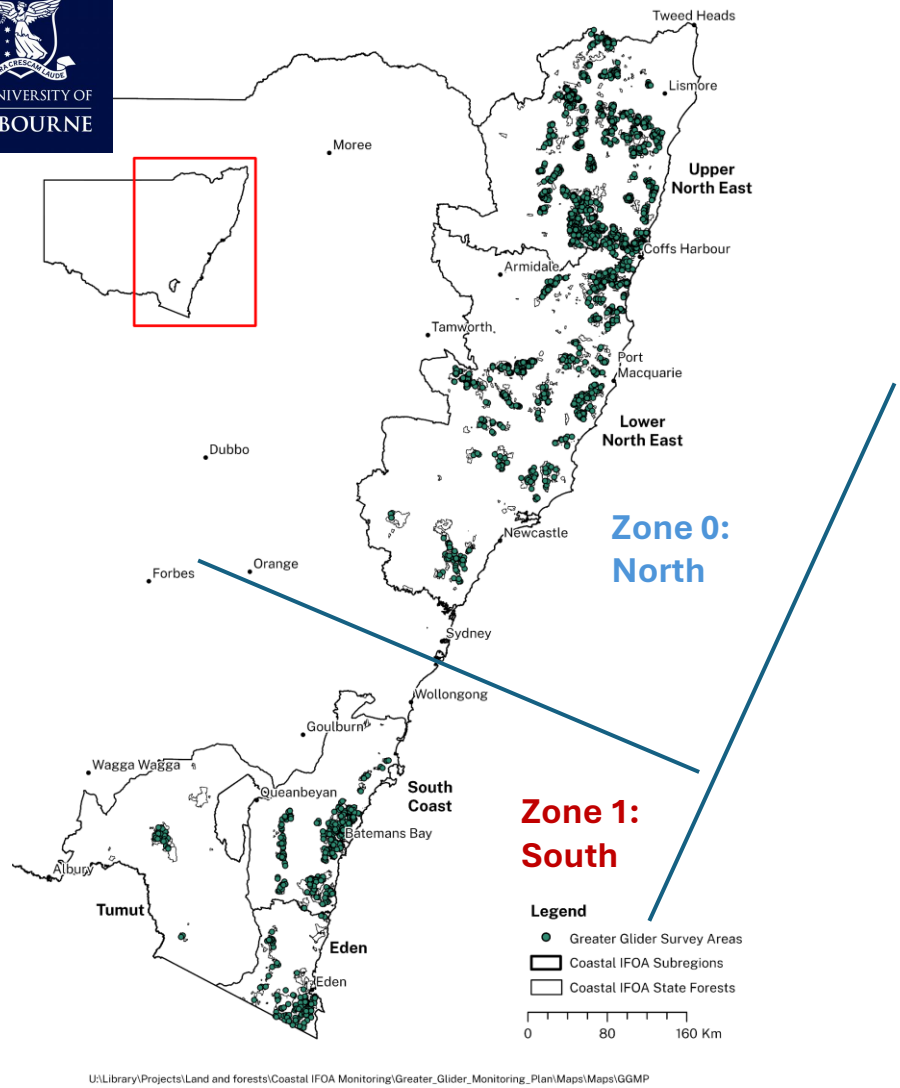
- 200 m
- 500 m
- 1000 m

- Fire
- Logging
- Combined

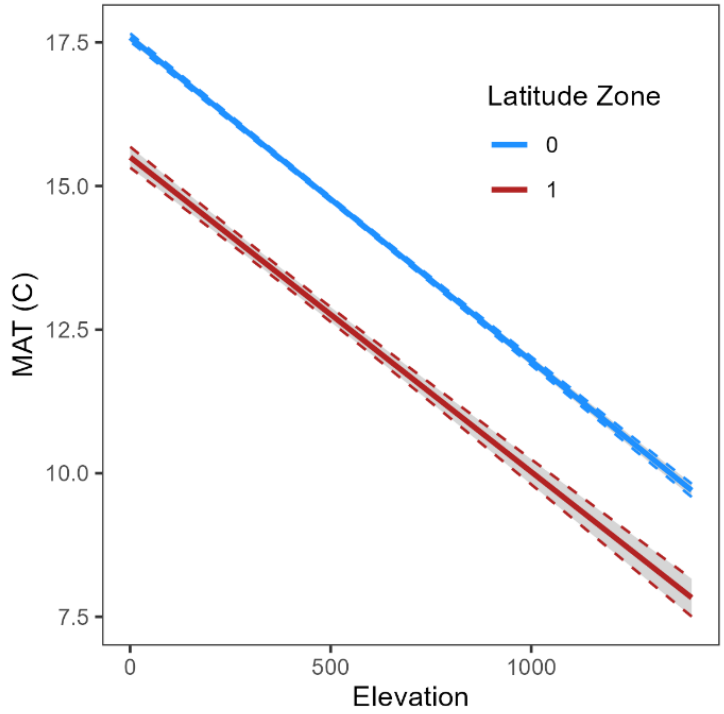




OFFICIAL



U:\Library\Projects\Land and forests\Coastal IFOA Monitoring\Greater_Glider_Monitoring_Plan\Maps\Maps\GGMP

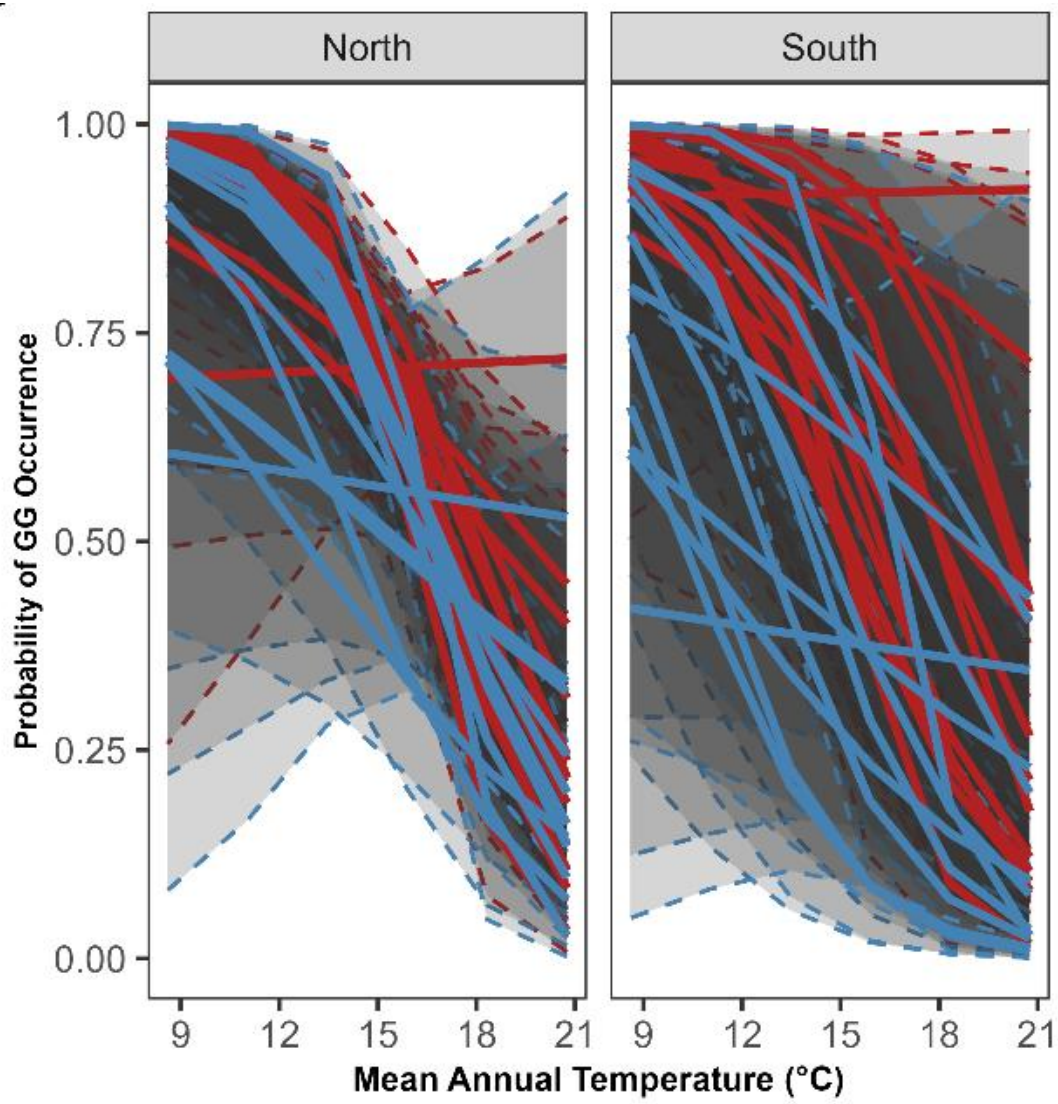




OFFICIAL

Year as a Random Effect

1998-2022 occupancy model

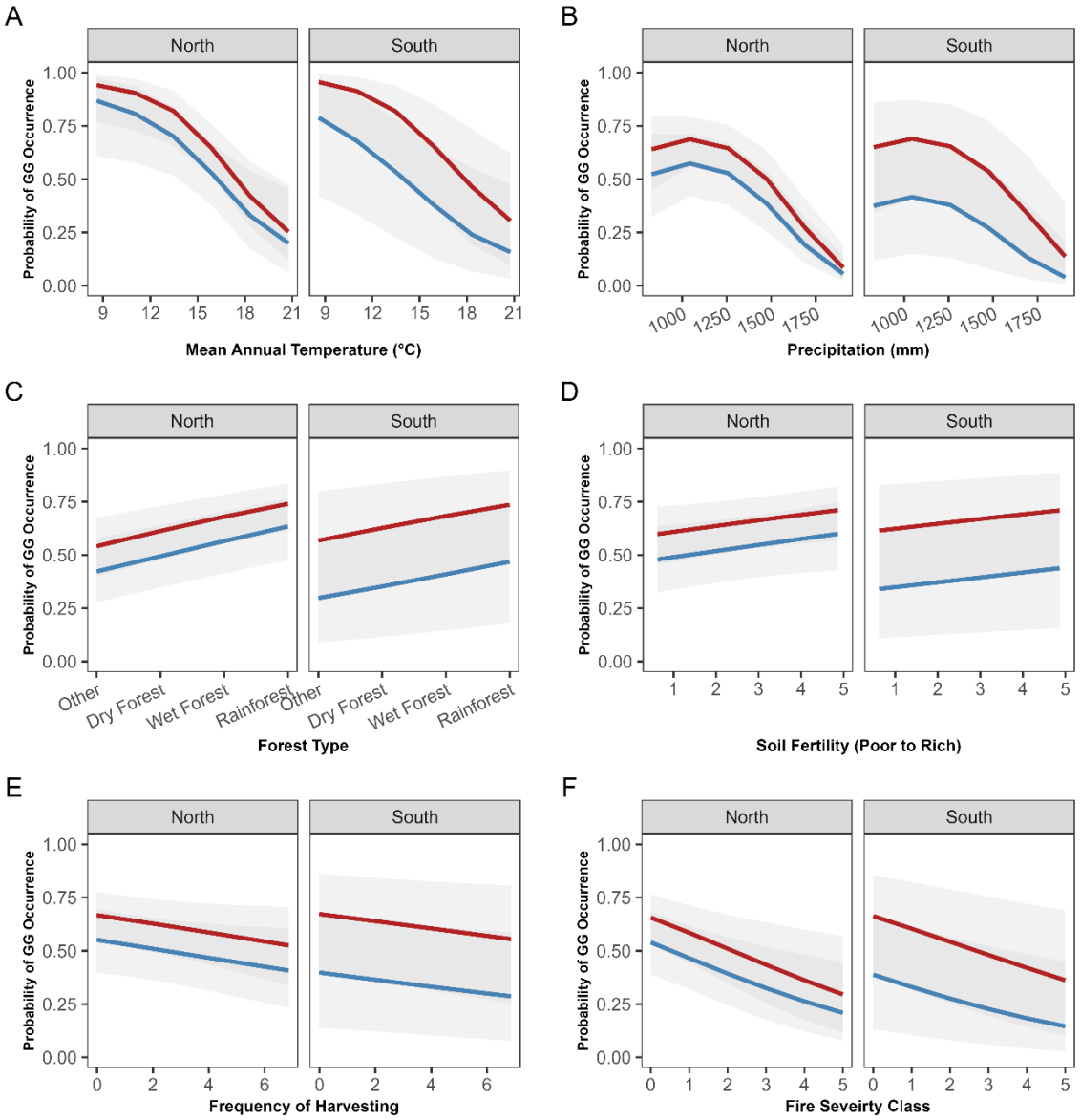


Red lines represent predictions for the years 1998 to 2010 and blue lines represent predictions for years 2011 and onwards.

OFFICIAL

OFFICIAL

Environmental factors in the 1998-2022 occupancy model

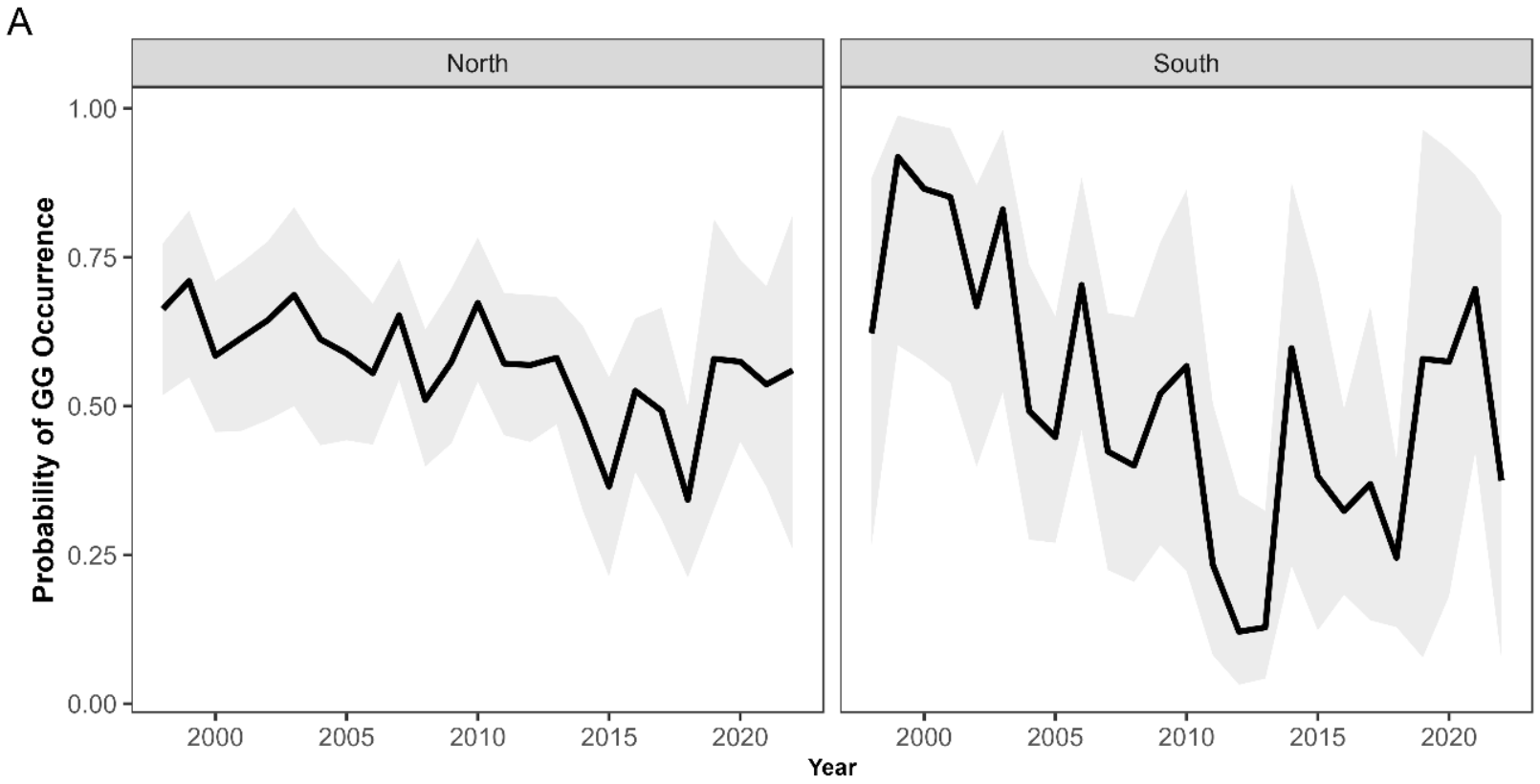


Red lines represent model predictions for years prior to and including 2010

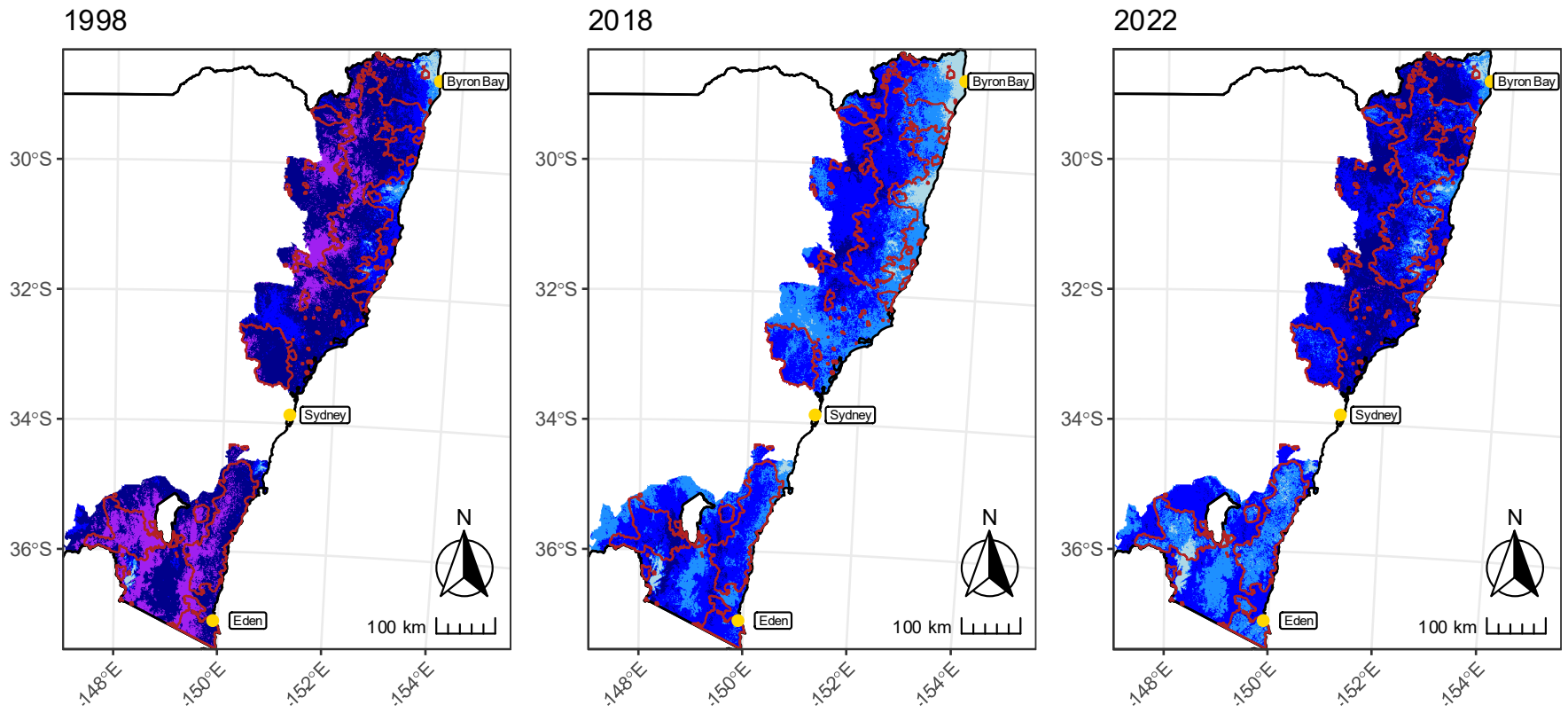
Blue lines represent model predictions for years post 2010

Model	Climate	Soil/ TWI /Forest Type	Fire	Logging	Latitude	Year
1998-2022	**67.7%**	1.4%	3.3%	0.5%	5.7%	21.5%

Predicted Occurrence

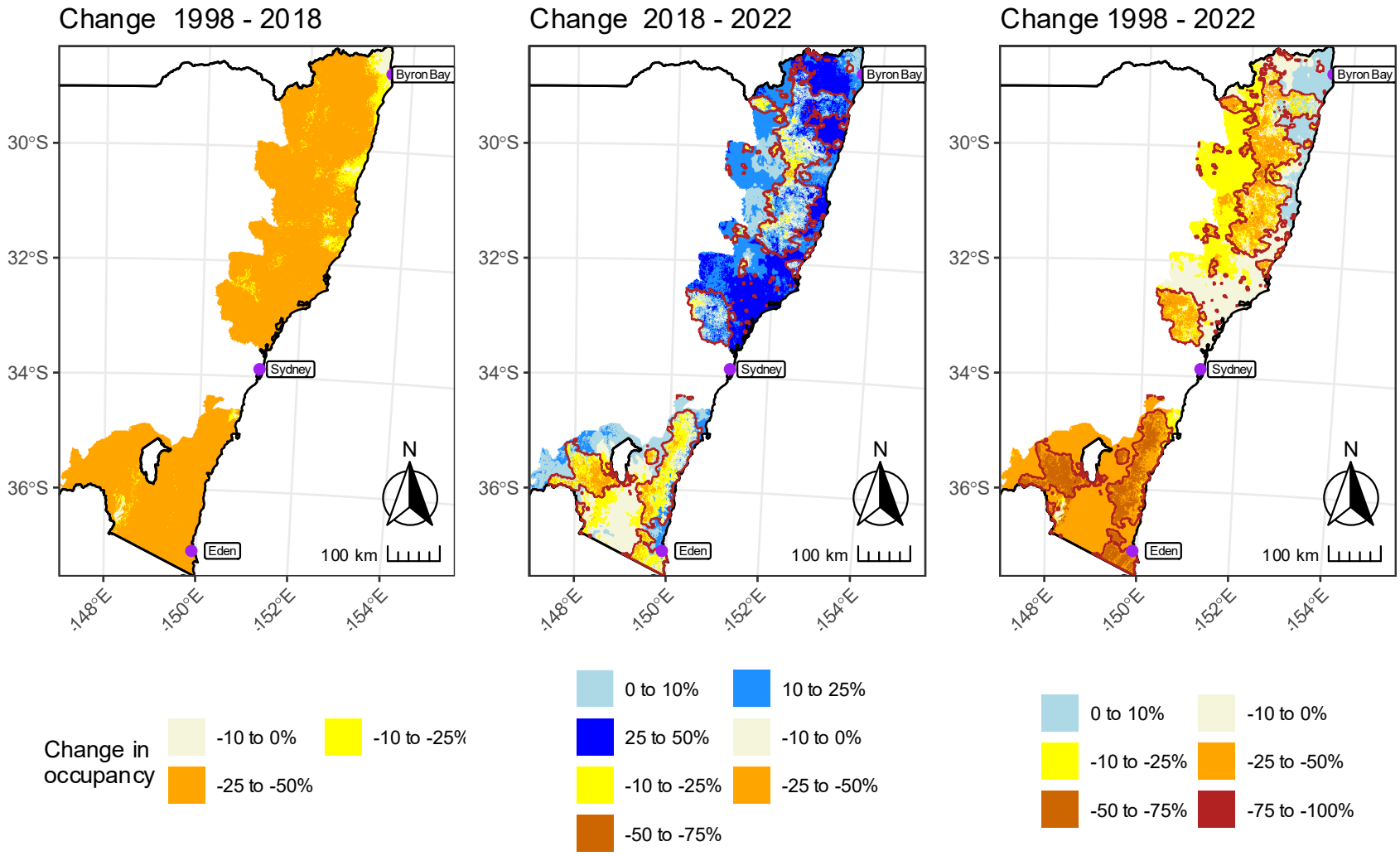


Modelled Occupancy



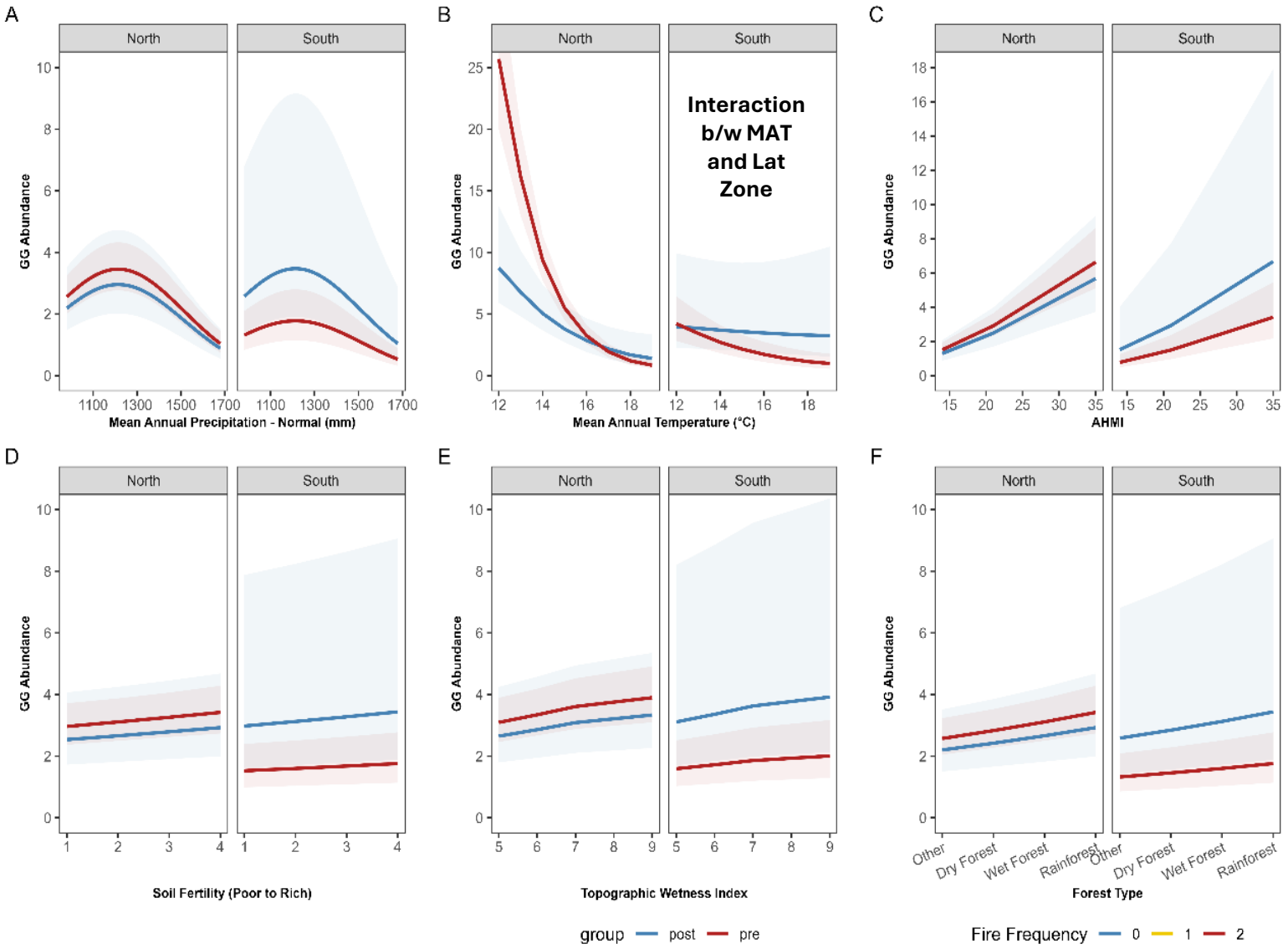
THE UNIVERSITY OF MELBOURNE

Modelled Change in Occupancy



OFFICIAL

Environmental factors in the 1998-2022 abundance model



Red lines represent model predictions for years prior to and including 2013

Blue lines represent model predictions for years post 2013

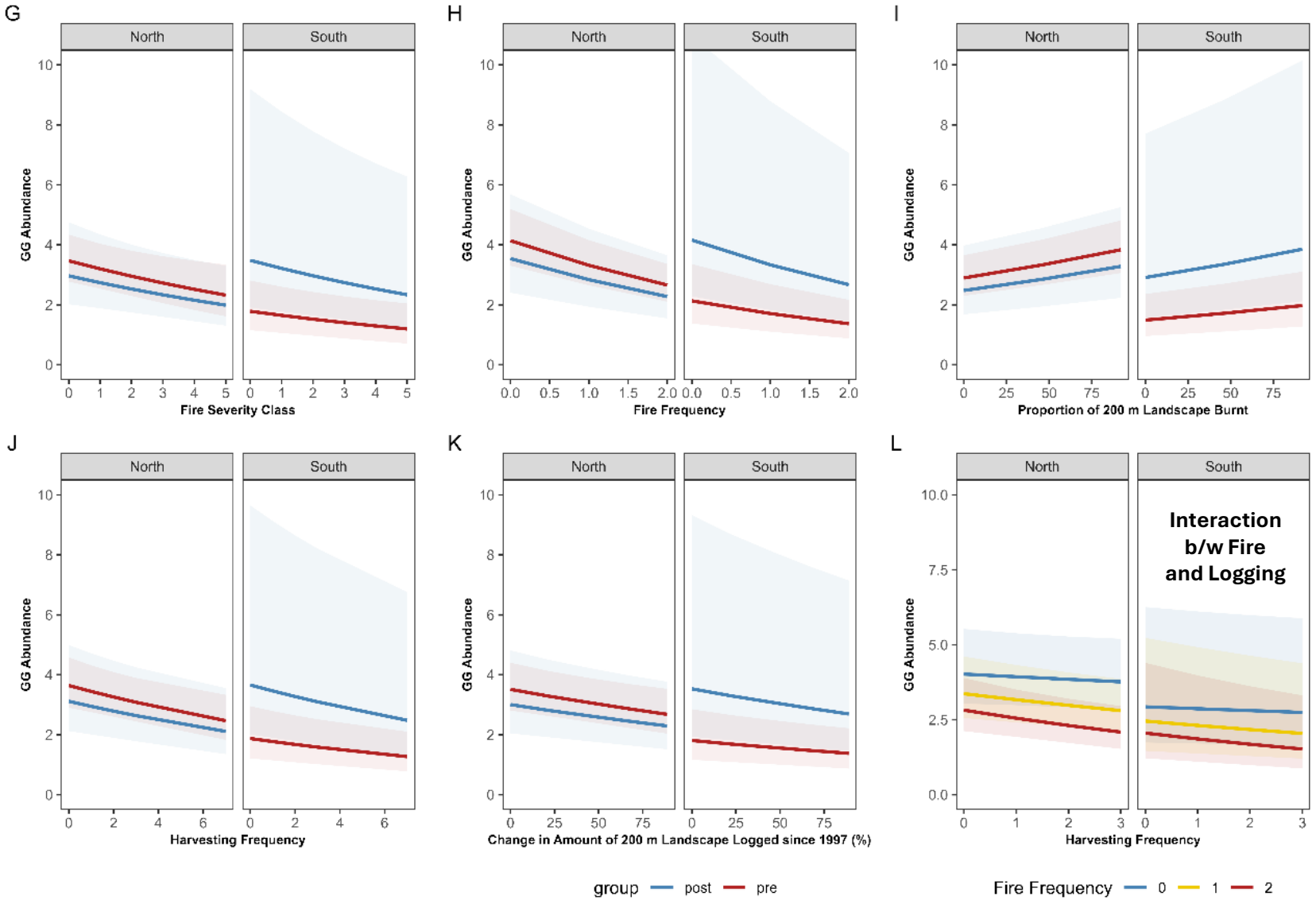
Model	Climate	Soil/ TWI/ Forest Type	Fire	Logging	Interactions	Latitude	Year
1998-2022	**60.8%**	1.4%	0.8%	0.4%	15.1%	2.9%	18.6%

THE UNIVERSITY OF MELBOURNE

OFFICIAL

OFFICIAL

Environmental factors in the 1998-2022 abundance model



Red lines represent model predictions for years prior to and including 2013

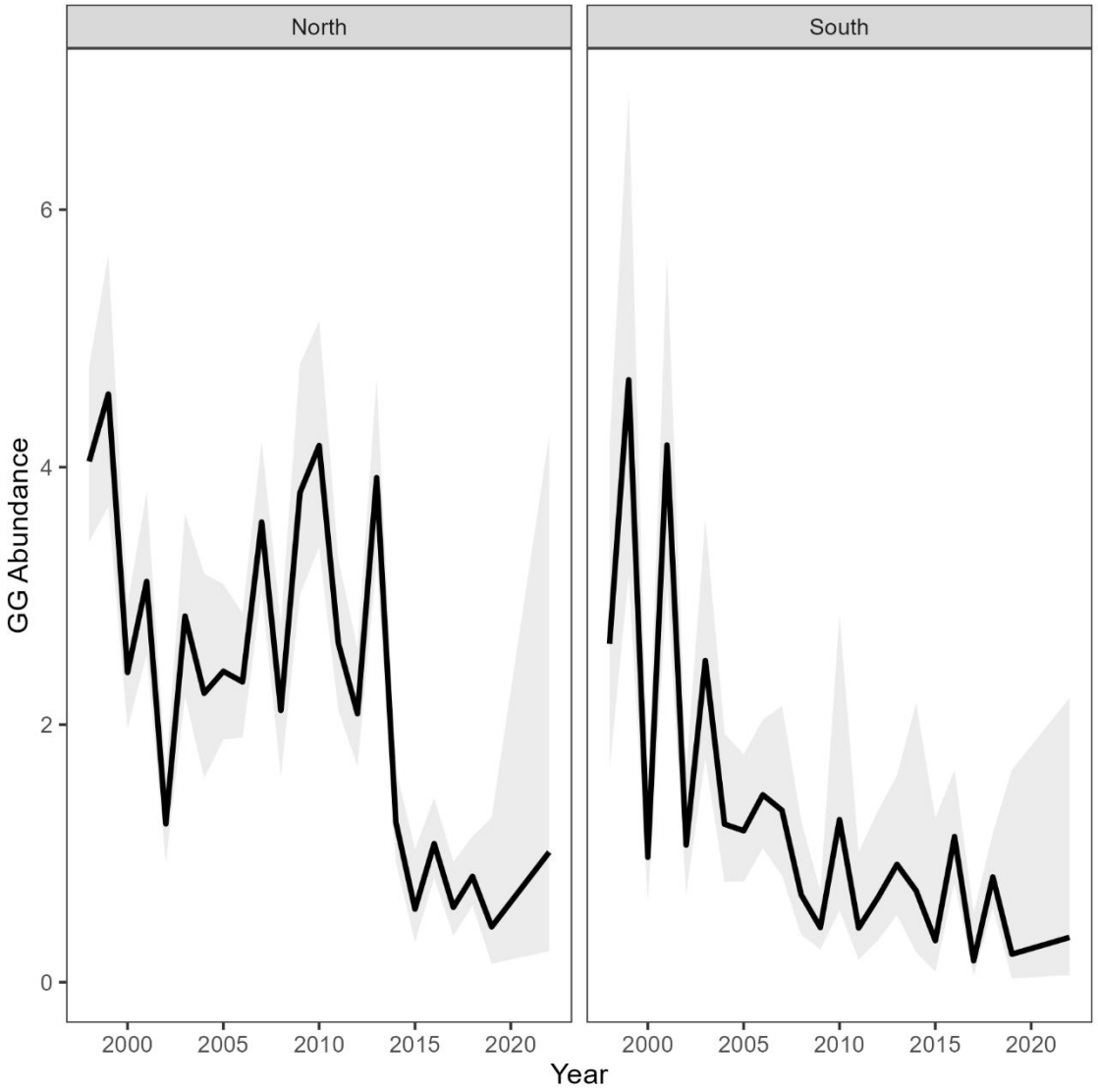
Blue lines represent model predictions for years post 2013

Model	Climate	Soil/ TWI/ Forest Type	Fire	Logging	Interactions	Latitude	Year
1998-2022	**60.8%**	1.4%	0.8%	0.4%	15.1%	2.9%	18.6%

THE UNIVERSITY OF MELBOURNE

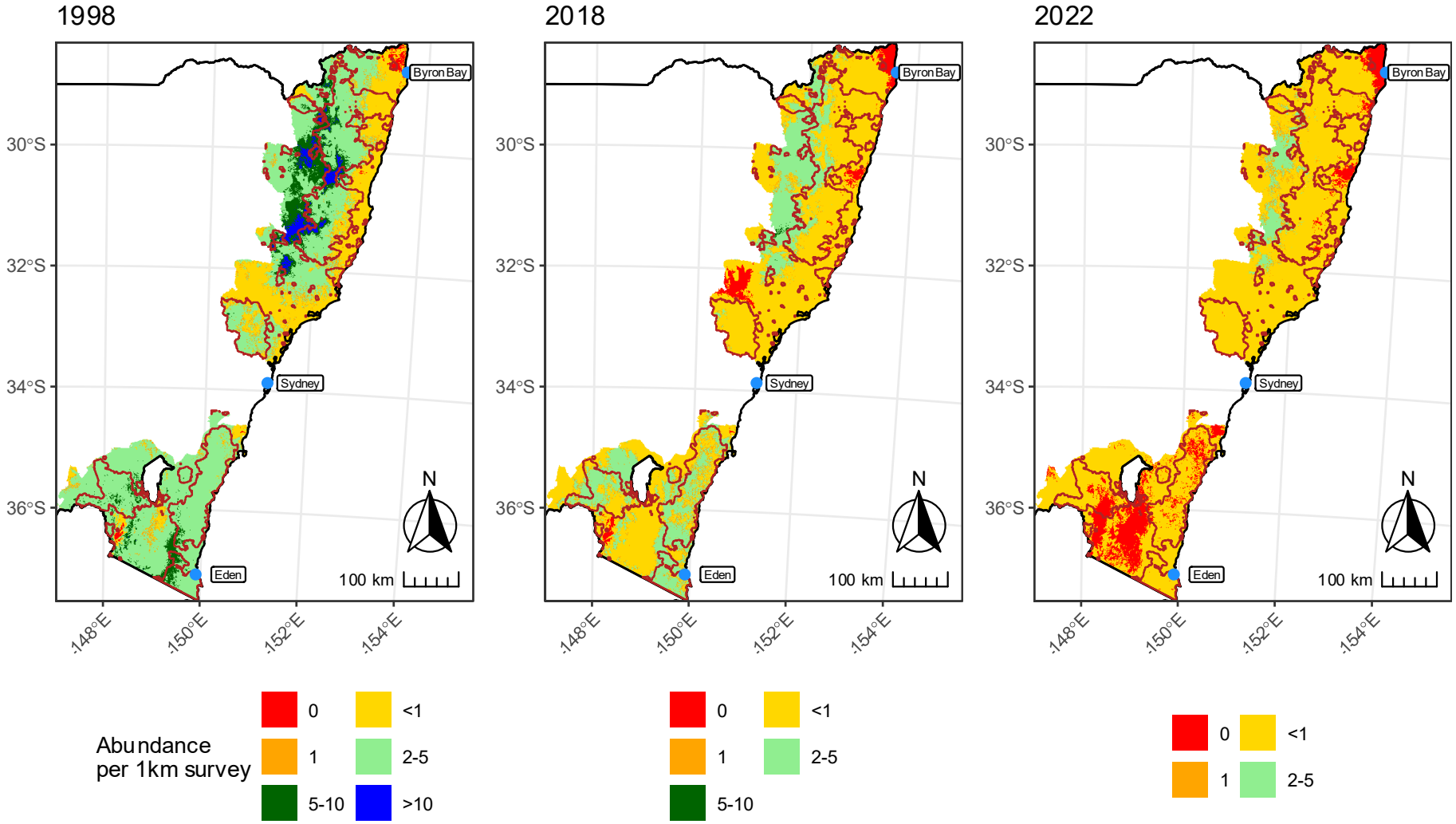
OFFICIAL

Predicted Abundance

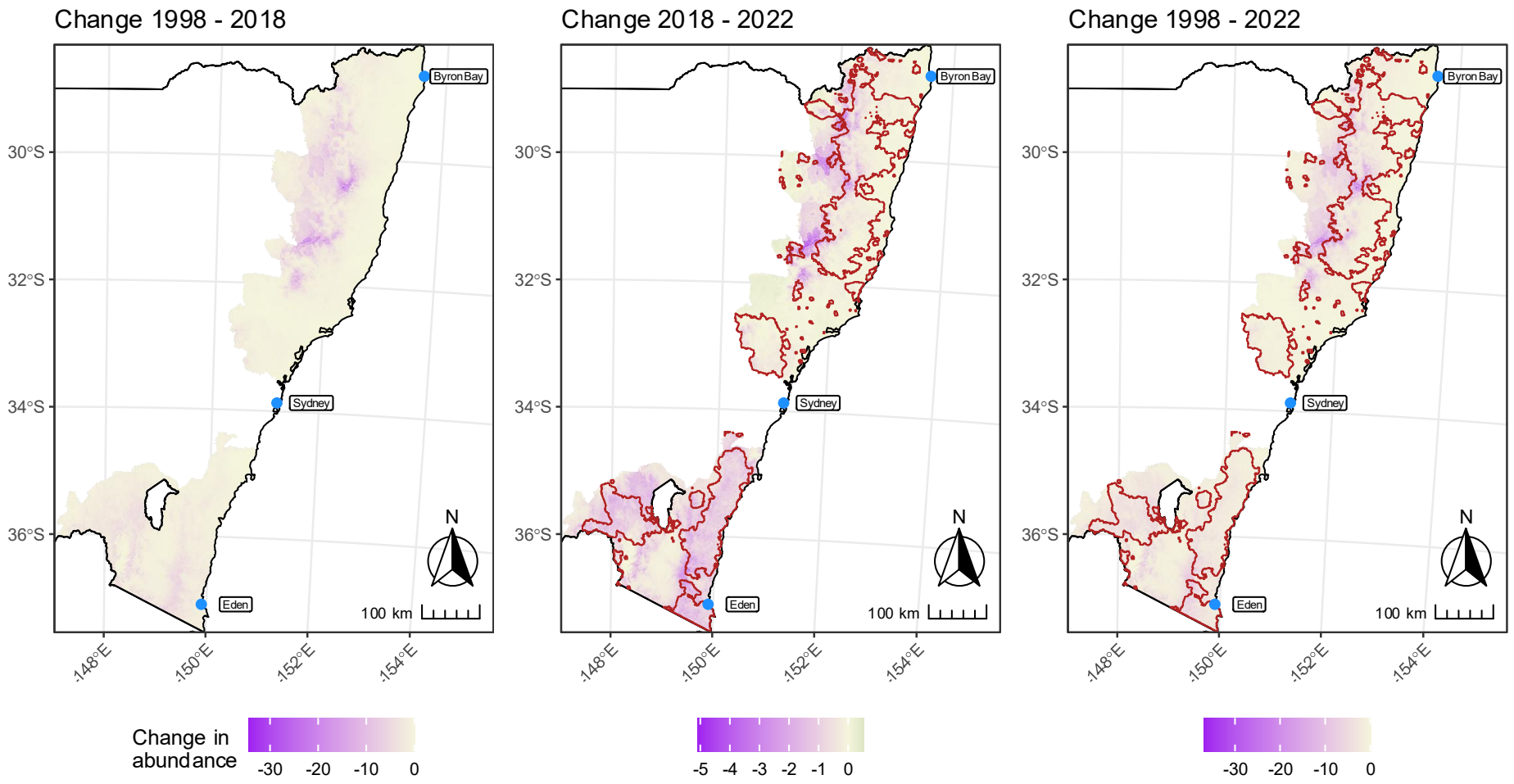




Modelled Abundance



Modelled Change in Abundance





Overall

- The decline in occurrence and abundance since 1998 aligns with past research
- Mean annual temperature was the most important factor for occupancy and abundance
- Strong effect of “Year” of survey and “Latitude” indicates that there are unresolved factors
- The effect of fire severity was three times greater on occupancy than abundance
- Fire frequency affected abundance not occupancy
- Increasing amount of the landscape burnt around sites had a positive effect on SGG abundance which could indicate a “fence effect”
- Strong evidence that historical harvesting frequency has had a small but significant negative effect on abundance

Limitations

- Models developed do not predict the impacts of subsequent harvesting
- Most sites assessed in this study were not riparian/ gully forests
- The unbalanced representation of post-fire (2019-2022) sites in the 1998-2022 model may have influenced model predictions
- Influence of occurrence of hollow bearing trees and the occurrence of suitable forage species not included



Final Thoughts

- Due to the need to consider “Year” as a random effect, monitoring is essential

- Long-term post-fire monitoring is needed to provide critical understanding of population recovery



Application

NSW GOVERNMENT Natural Resources Commission

Follow us on Linkedin

Search

HOME | ABOUT US | CURRENT WORK | COMPLETED WORK | PUBLICATIONS | OTHER INFO

FMIP Home > CIFOA Home

Coastal IFOA monitoring program


Forest health


Biodiversity


Water quality


Research and evaluation


Wood supply



Reporting

The monitoring program

We are independently overseeing a monitoring program with the NSW Forest Monitoring Steering Committee to ensure the ongoing effectiveness of the Coastal Integrated Forestry Operations Approval (Coastal IFOA) in achieving its objectives and outcomes.

Forests

- COASTAL IFOA OPERATIONS POST BUSHFIRES
- COASTAL IFOA MONITORING
- FOREST DIEBACK
- FOREST MONITORING AND IMPROVEMENT PROGRAM
- KOALA RESEARCH
- PRIVATE NATIVE FORESTRY MONITORING
- RFA MER PLAN

Read Latest News

Read Latest Reports



Project B3: Long term Greater Glider monitoring

Project B3: Long term Greater Glider monitoring

The site-specific biodiversity condition for Greater Gliders in the Coastal IFOA region requires the CIFOA monitoring program to monitor the effectiveness of site-specific conditions to protect Greater Glider populations in mapped areas.

The program has engaged Dr Rod Kavanagh, Adjunct Associate Professor at Southern Cross University to design and propose a plan to meet the new requirements. For over four decades, Dr Kavanagh has worked as a senior wildlife research scientist and forest ecologist across government, the private sector and not-for-profit organisations. He has published in peer reviewed scientific journals, including research on Greater Gliders, other arboreal marsupials and large forest owls in south eastern NSW forests.

In designing a plan, Dr Kavanagh will work closely with a cross-agency technical working group including our staff, Environment Protection Agency, Forestry Corporation of NSW, Department of Climate Change, Energy, the Environment and Water, and Department of Primary Industries.

Thank you

NSW GOVERNMENT

THE UNIVERSITY OF MELBOURNE

LANDSCAPE & FOREST DYNAMICS GROUP

Contact:
Craig: craign@unimelb.edu.au
Ben: benjamin.wagner@unimelb.edu.au